



PRODUCT DATASHEET

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Product:	Micro Servo 9g High Torque 5kg.cm
Part No:	DS9G-270 (No Feedback)
Part No:	DS9G-270-FB (With Feedback)

1. DESCRIPTION

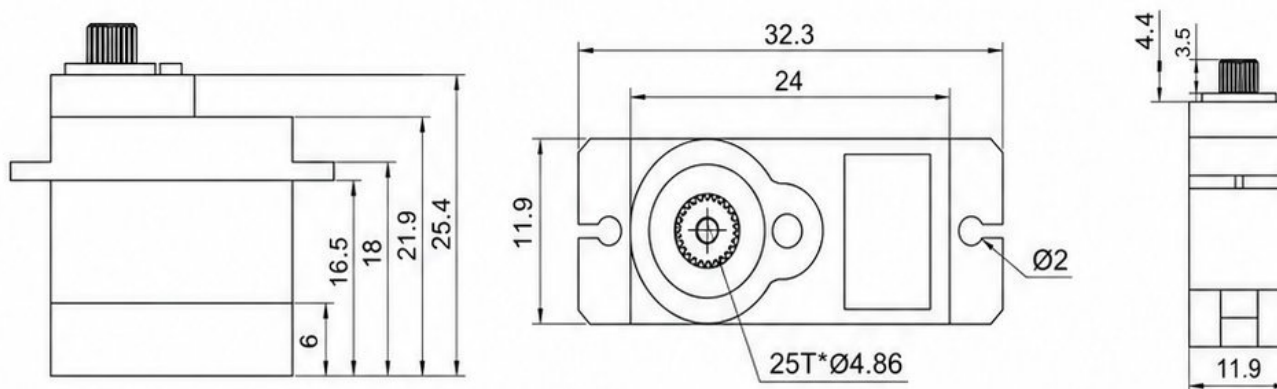
The DS9G-270-FB Micro Servo offers 5kg.cm torque at 8.4V, making it powerful for various applications. It features a coreless motor, double bearing design and operates with a 270-degree range, providing smooth and precise movement.





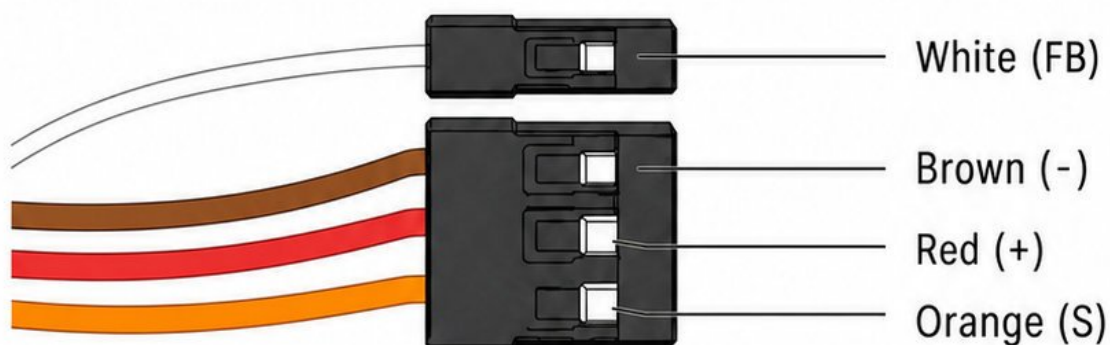
TECHNICAL DETAILS

2. DIMENSIONS



3. CONNECTOR PINOUTS

Type: JR (3+1 pin), 2.54 mm





TECHNICAL SPECIFICATIONS

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4. ENVIRONMENTAL CONDITIONS

No.	Item	Specification
1-1	Storage Temperature Range	-30°C to 80°C
1-2	Operating Temperature Range	-15°C to 70°C
1-3	Operating Voltage Range	4.8–8.4 V

5. MECHANICAL SPECIFICATIONS

No.	Item	Specification
2-1	Size	24 × 11.8 × 21.9 mm
2-2	Weight	13 g
2-3	Gear Ratio	410
2-4	Bearing	Double bearing
2-5	Connector Wire	170 ± 5 mm
2-6	Motor	3-pole(k)
2-7	Waterproof Performance	NO

6. ELECTRICAL SPECIFICATIONS

No.	Item	6 V	7.4 V	8.4 V
3-1	Idle Current (at stopped)	4 mA	5 mA	6 mA
3-2	Operating Speed (at no load)	0.14 sec/60°	0.12 sec/60°	0.10 sec/60°
3-3	Stall Torque (at locked)	3.5 kg-cm	4 kg-cm	5 kg-cm
3-4	Stall Current (at locked)	0.6 A	0.8 A	1.0 A

7. CONTROL SPECIFICATIONS

No.	Item	Specification
4-1	Control System	PWM (Pulse Width Modulation)
4-2	Pulse Width Range	500–2500 μsec
4-3	Neutral Position	1500 μsec
4-4	Operating Angle	180° or 270° (at 500–2500 μsec)
4-5	Dead Band Width	3 μsec
4-6	Operating Frequency	50–330 Hz
4-7	Rotation Direction	Counterclockwise (at 500–2500 μsec)



8. PWM SIGNAL

